MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, K6 OPERATION AND SEQUENCE

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-6O898 comprise the complete requirements for the K6 Operation and Sequence Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue In effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10554185	Electronic Component Assembly, KU Operation l nd Sequence

PACKAGING DATA SHEET

10554185	Packaging of Circuit Card Assembly, K6 Oper-ation and Sequence

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554185 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-60898.

3.3 Environmental service conditions. See MIL-C-60898.

3.3.1 Performance at temperature extremes. For the conditions of 3.3.2 of MIL-C-6089 8, the foil owing requirements shall be met for 3.4.

a) Logic "1" output. An output logic "1" shall be greater than +2.5 vdc but less than +4.4 vdc.

b) Logic "0" output. An output logic "O" shall be greater than +0.0 1 vdc but less than +0.4 vdc.

c) Pin 21 output. The output voltage of pin 21 shall be either +20 ± 1.5 vdc or greater than +0.05 vdc but less than +1.2 vdc, respectively.

d) Pin 6 output. The output voltage of pin 6 shall be either +20.5 ± 1.5 vdc or greater than +0.01 vdc but less than +1.2 vdc, respectively.

e) Pin 23 output. The output voltage of pin 23 shall be either +18.0 ± 1.0 vdc or greater than +0.01 vdc but less than +1.2 vdc, respectively.

3.4 Electrical performance

3.4.1 Conditions of operation. The requirements of 3.4.2 and 3.4.3 shall be met with the terminations to the board as shown in Figure 1 unless otherwise specified. A logic "O" input shall be +0.50 ± 0.01 vdc and a logic "1" input shall be +2.80 ± 0.01 vdc.

3.4.2 Logic outputs. An output logic "l" shall be greater than +2.8 vdc but less than +4.4 vdc. An output logic "O" shall be greater than +0.01 vdc but less than +0.4 vdc. Pin 28 shall be ground.

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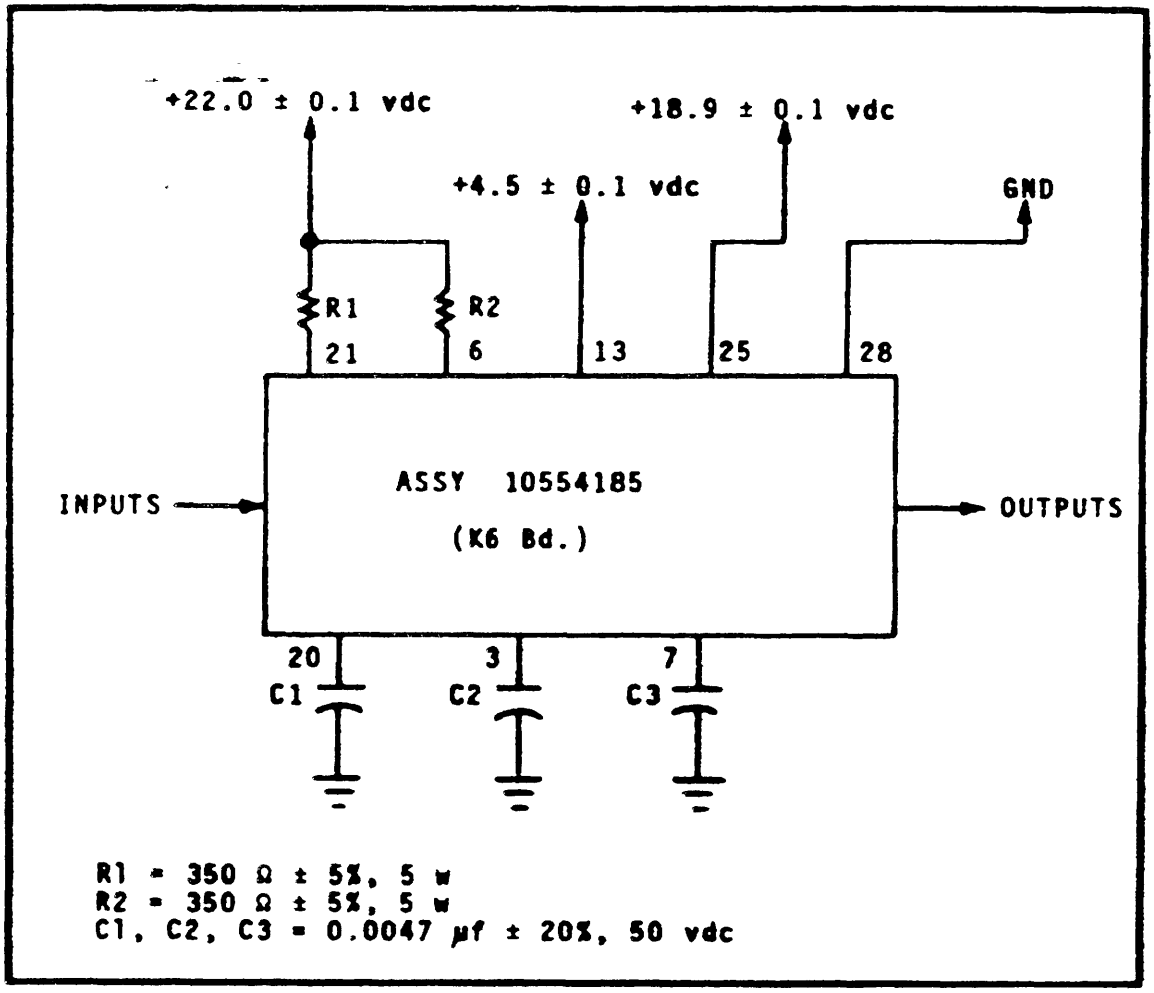


Figure 1

3.4.2.1 Output pin 9. With the inputs applied as specified in Table I, the output at pin 9 shall be either an output logic "1" or an output logic "0" as specified in Table 1.

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Table I

Input pins			Output pin
15	16	17	9
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

3.4.2.2 Output pin 23. With a logic input "1" applied to pin 19, the voltage at pin 23 shall be +18.4 ± 0.5 vdc. With a logic input "O" applied to pin 19, the voltage at pin 23 shall be greater than +0.01 vdc but less than +1.2 vdc.

3.4.2.3 Output pin 6. The voltage between pin 6 (+) and pin 28 shall be greater than +0.01 vdc but less than +1.2 vdc for the conditions of 3.4.1. With pin 13 grounded, the voltage between pin 6 and pin 28 shall be +21.5 ± 0.5 vdc.

3.4.3 Sequence and timing. With the inputs applied in the sequence specified in Table II, there shall be the outputs specified in Table II at pin 7 and at pin 21.

Table II

Step	Input pins					Output pins	
	15	16	17	18	19	7	21
1	1	1	1	**	0	0	+0.05 vdc to +1.2 vdc
2	0	1	1	1	0	1*	+0.05 vdc to +1.2 vdc
3	0	1	1	1	1	1	+21.5 ± 0.5 vdc
4	1	1	1	1	0	0	+0.05 vdc to +1.2 vdc
5	1	1	1	**	0	0	+0.05 vdc to +1.2 vdc

* Logic "1" delayed 500 ± 25 milliseconds from the time pin 15 becomes a logic "O".

** Apply a "O" for 1 to 5 seconds and then apply a "1".

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3.4.4 Capacitor C1. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 25 (+) and pin 28, the voltage out between pin 25 (+) and pin 28 shall be l s shown in Figure 3.

3.4.5 Capacitor C2. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 13 (+) and pin 1, the voltage out between pin 13 (+) and pin-l shall be as shown in Figure 3.

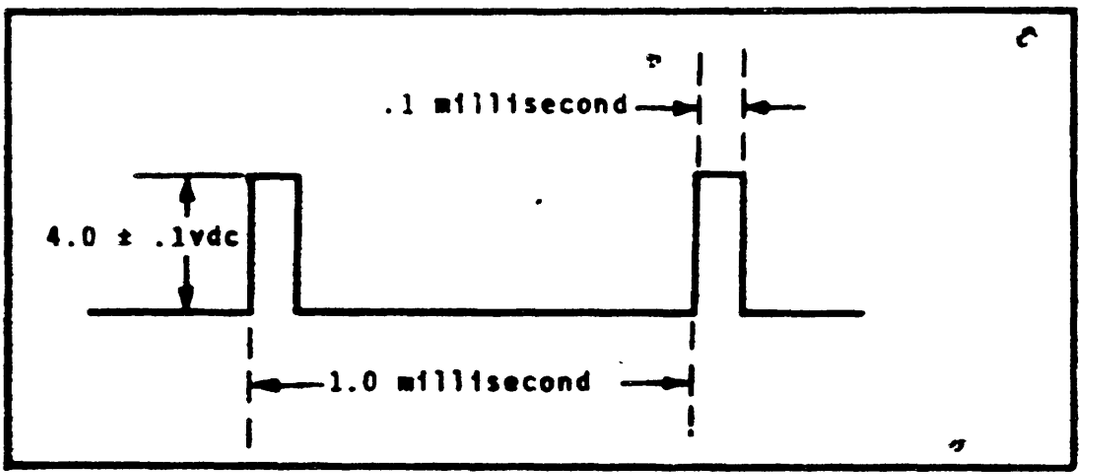


Figure 2

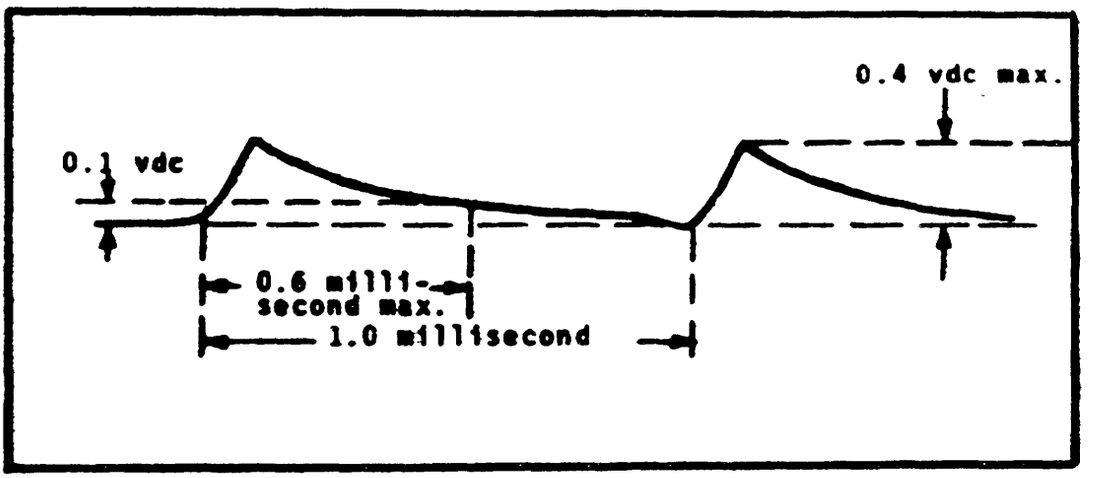


Figure 3

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. As specified in MIL-C-60898.

4.4 Functional tests. The following characteristics shall be 100 percent tested in accordance with the applicable test method and MIL-C-60898.

	Characteristic	Requirement	Procedure
100	Conditions of Operation	3.4.1	4.5.2.1
101	Logic outputs	3.4.2	4.5.2.2
102	Output pin 9	3.4.2.1	4.5.2.2.1
103	Output pin 23	3.4.2.2	4.5.2.2.2
104	Output pin 6	3.4.2.3	4.5.2.2.3
105	Sequence and timing	3.4.3	4.5.2.3
106	Capacitor, C1	3.4.4	4.5.2.4
107	Capacitor, C2	3.4.5	4.5.2.5

4.4.5.1 Test equipment. In conjunction with 4.4.1, the following test l quipment shall be utilized in the Performance of the l pplicable tests l s specified in 4.5.

	Item	Description
1.	POWER SUPPLY	22.0 ± 0.1 vdc, 500 ma or greater. Ripple less than 100 mv rms.
2.	POWER SUPPLY	4.5 ± 0.1 vdc, 500 ma or greater. Ripple less than 100 mv rms.
3.	POWER SUPPLY	18.9 ± O.1 vdc, 500 ma or greater Ripple less than 100 mv rms.
4.	POWER SUPPLY	0.50 ± 0.01 vdc, 50 ma or greater. Ripple less than 5 mv rms.

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5.	POWER SUPPLY	2.80 ± 0.01 vdc, 50 ma or greater. Ripple less than 5 mv rms.
6.	DIGITAL VOLTMETER	With accuracy and resolution of ± 0.2 percent or better from 0.0090 to 99.0 vdc. Input impedance 10 megohms or greater on all voltage scales.
7.	OSCILLOSCOPE	Voltage and time accuracy and resolution of ± 5 percent or better. Rise time 100 nanoseconds or less; sweep periods from 1 microsecond to 1 second per centimeter. Amplitude scales from 0.01 to 10 vdc per centimeter, automatic triggering.
8.	ASSORTED COMPONENTS	Two resistors, 350 ohm ± 5 percent, 5 watt. Three CAPACITORS, 0.0047 uf ± 20 percent, 50 vdc. One resistor 100 ohms ± 10 percent, ½ watt.
9.	PULSE GENERATOR	Frequency-1 kilohertz minimum. Duty factor -10 percent minimum. Amplitude-minimum of 4 volts into a 200 ohm load. Pulse rise and fall times-less than 0.2 microseconds.

4.5 Test methods and procedures

4.5.1 Environmental. As specified in MIL-C-60898 except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4.5.2 Electrical tests

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4.5.2.1 Conditions of operation. Unless otherwise specified, all subsequent tests are to be performed with the conditions of operation specified in 3.4.1, Referring to Figure I, connect R1,R2, C1, C2 and C3, using the ASSORTED COMPONENTS, item 8 of 4 4.1.1. Connect the POWER SUPPLIES, items 1, 2, and 3 of 4.4.1.1, to supply +22.0 3 0.1 vdc, +4.50 ± 0.01 vdc and +18 9 ' 0.1 vdc, respectively, with their negative terminals tied to GROUND as shown in Figure 1. When required during subsequent test, use the positive output of the POWER SUPPLY, item 4 of 4.4.1.1, as logic "O" and the positive output of the POWER SUPPLY, item 5 of 4.4.1.1, as logic "l". Connect the negative terminals of these two supplies to GROUND of Figure I.

4.5.2.2 Logic outputs. In the following tests, use the DIGITAL VOLTMETER item 6 of 4.4.1.1, to verify that the logic outputs are'within the tolerances specified in 3.4.2.

4.5.2. 2.1 Output pin 9. Referring to Table I, apply logic inputs as shown for pins 15, 16 and 17. Verify that the output voltage on pin 9-conforms to the requirements of Table I and 3.4.2.1 for each combination of inputs on pins 15, 16 and 17.

4.5.2.2.2 Output pin 23. Connect the DIGITAL VOLTMETER, item 3 of 4.4.1.1 to output pin 23. Verify an Output of 18.4 ± 0.5 vdc with a logic "1" applied to-pin 19. Verify an output of +0.01 vdc to +1.2 vdc with a logic "O" applied to pin 19.

4.5.2. 2.3 Output pin 6. Using the DIGITAL VOLTMETER, item 3 of 4.4.1.1, verifyy an output between pin 6 (+) and pin 28 Of +0.01 vdc to +1.2 vdc Connect pin 13 to pin 28 (GROUND) and verify an output between pin 6 (+) and pin 28 of 21.5 ± 0.5 vdc.

4,5.2.3 Sequence and timing. Referring to Table II, apply logic inputs as shown for pins 15, 16, 17, 18 and 19 for each step of Table II. Connect the OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 7 (+) and in 1. Connect the OSCILLOSCOPE trigger input between pin 15 (+) and pin 1. Adjust the OSCILLOSCOPE to trigger when pin 5 goes from a logic "1" to a logic "0". Verify that the output voltage on pins 7 and 21 conform to the requirements of Table II and 3.4.3 for each specified combination of inputs on pins 15, 16, 17, 18 and 19. Using the OSCILLOSCOPE, confirm STEP 2 of Table II by vertfying that pin 7 goes from logic "O" to logic "1" within 500 ± 25 milliseconds after pin 15 is changed from a logic "1" to a logic "0", with all other input pins at their specified logic level.

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4.5.2.4 Capacitor C1. Connect the PULSE GENERATOR, item 9 of 4.4.1.1, thru the 100 OHM RESISTOR, item 8 of 4.4.1.1, and connect channel A vertical input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 25 (+) and pin 28 per Figure 4. Adjust the PULSE GENERATOR to produce the pulse of Figure 2, Connect the OSCILLOSCOPE channel B vertical input, per Figure 4, and verify the waveform of Figure 3.

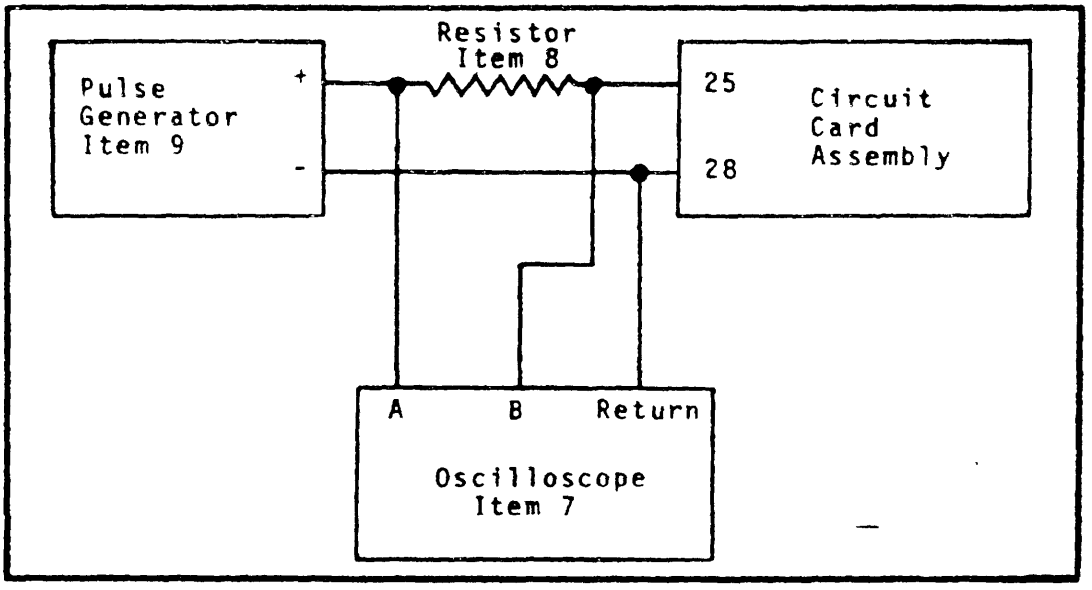


Figure 4

4.5.2.5 Capacitor C2. Connect the PULSE GENERATOR, item 9 of 4.4.1.1, thru the 100 OHM RESISTOR, item 8 of 4.4.1.1, and connect channel A vertical input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 13 (+) and pin 28 per Figure 5. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B vertical input, per Figure 5, and verify the waveform of Figure 3.

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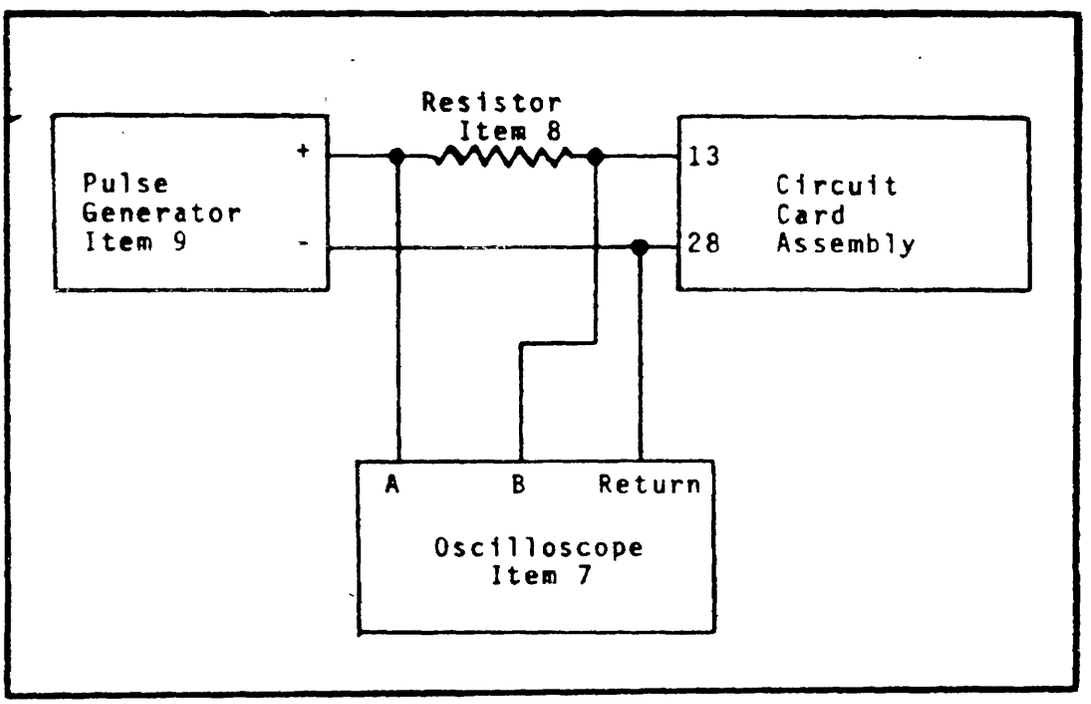


Figure 5

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554185.

6. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

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